Unrolling Nonnegative Matrix Factorization With Group Sparsity for Blind Hyperspectral Unmixing

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***Abstract*— Deep neural networks have shown huge potential in hyperspectral unmixing (HU). However, the large function space increases the difficulty of obtaining the optimal solution with limited unmixing data. The autoencoder-based blind unmixing methods are sensitive to the hyperparameters, and the optimal solution can be difficult to obtain. Algorithm unrolling, which integrates deep learning and iterative algorithms, can shrink the search space and improve the efficiency of obtaining optimal results. Based on this, a model-driven deep neural network named the group sparsity regularized unmixing unrolling (GSUU) network, which unrolls a regularized matrix factorization objective function for blind HU, is proposed in this article. Based on the nonnegative matrix factorization (NMF) optimization rules, the GSUU network contains two subnetworks—the A-Block and the S-Block—for alternately and iteratively estimating the optimal endmember spectra and abundance maps. The GSUU method incorporates the spatial group sparsity prior to the abundances, i.e., the fact that spatially adjacent mixed pixels share similar sparse abundances, into a deep unrolling network. The experimental results obtained with both synthetic and real hyperspectral data illustrate that the proposed algorithm can obtain a superior accuracy, compared to the other state-of-the-art unmixing algorithms.**

***Index Terms*— Deep neural network, deep unrolling, group sparsity, hyperspectral unmixing (HU).**

NOMENCLATURE

$\mathbf{X} \in \mathbb{R}^{L\times N}$	Hyperspectral image.
$X^1, \ldots, X^P$	Superpixels sliced by simple linear iterative clustering (SLIC).
$\mathbf{W}^p \in \mathbb{R}^{M\times N}$	Superpixelwise weight matrix.
$\mathbf{A}_{k+1} \in \mathbb{R}^{L\times M}$	Endmember matrix output of the kth layer.
$\mathbf{S}_{k+1} \in \mathbb{R}^{M\times N}$	Abundance matrix output of the kth layer.
$\eta(\cdot)$	Activation function.

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I. INTRODUCTION

HYPERSPECTRAL images contain hundreds of narrow spectral bands, which can be used for analyzing the detailed information of materials [1], [2], [3], [4]. However, due to the low spatial resolution and the multiple scattering of light, mixed pixels that contain more than one pure material (endmember) are common in hyperspectral imagery, which affects the material identification accuracy for hyperspectral data analysis.

Hyperspectral unmixing (HU) is an effective way to address the mixed pixel problem, where each pixel is decomposed into a set of endmembers and their corresponding abundances. In general, HU is an ill-posed blind source separation problem, which is nonconvex and may not guarantee a unique solution since both the endmembers and abundances need to be estimated from the hyperspectral imagery [5]. Image priors are essential for blind HU [6], e.g., convex geometry [7], independency [8], [9], [10], [11], nonnegativity [12], spatial correlation [13], [14], and sparsity [15], [16], [17], to increase the stability of HU.

The convex geometry prior is an important prior, which assumes that endmembers tend to be located at the vertices of the simplex enclosing the hyperspectral data. The traditional HU methods, e.g., N-FINDR [18], orthogonal subspace projection (OSP) [19], and vertex component analysis (VCA) [20], sequentially extract endmembers by using this prior [21], [22], and then follow this up with fully constrained least squares (FCLS) for the abundance estimation [12]. Blind HU can also be modeled as a nonnegative matrix factorization (NMF) problem by iteratively decomposing a nonnegative observation matrix into two low-rank nonnegative matrices, i.e., endmember and abundance matrices. NMF is ideal for blind HU due to its nonnegative assumption and mathematical basis. Additional image priors can also be integrated to further improve the unmixing accuracy [23], [24], e.g., the convex geometry prior used in minimum-volume-constrained NMF (MVCNMF) [25], the convex geometry and sparsity priors used in the sparsity promoting iterated constrained endmember (SPICE) algorithm [26], the sparsity prior used in $L_{1/2}$ regularized NMF ($L_{1/2}$NMF) [27], and the spatial correlation and sparsity prior used in spatial group sparsity regularized NMF (SGSNMF) [28].

Recently, unsupervised deep learning has been used to solve blind HU problems and has achieved good results.

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Autoencoders (AEs) are the main benchmark deep learning models for blind HU, which are aimed at learning a low-dimensional representation of the input hyperspectral data in an unsupervised and data-driven manner [29]. The encoder and the decoder model the unmixing and mixing procedures, respectively. Various AEs, e.g., the sparse AE [30], [31], the denoising AE [32], the stacked AE [33], [34], the variational AE [33], the convolutional AE [35], [36], and the adversarial AE [37], have been introduced to and integrated with unmixing constraints for blind HU. Specifically, to improve the robustness and stability of blind HU, the untied denoising autoencoder with sparsity (uDAS) [32] was proposed by introducing a denoising AE and associated with an L_{21}-norm sparsity constraint for regularizing the abundances. To promote the self-reconstruction performance of AEs, Ozkan et al. [30] proposed EndNet by using the spectral angle distance (SAD) as the loss function of the decoder, and Min et al. [31] added the Wasserstein distance in the loss function for modeling the distribution similarity between the input and output data. In order to avoid outliers, Su et al. [34] used the stacked AE to detect the outliers to reduce the interference. Su et al. [33] also combined the stacked AE and variational AE to obtain good initial spectral signatures for blind HU. Considering the spatial correlation, Palsson et al. [35] introduced a convolutional AE for jointly extracting spatial and spectral features, and Gao et al. [36] introduced the cycle-consistent AE architecture, i.e., two cascaded convolutional AEs as a cycle, to further learn the detailed information. Jin et al. [37] used the discriminative part of the adversarial AE to consider the intrinsic spatial information and obtain a prior distribution to guide the network output toward meaningful results.

In general, the AE-based HU methods can effectively model the unmixing procedure in a data-driven manner, while various physical constraints, e.g., the nonnegativity, the sparsity, or the sum-to-one constraint of the abundances, can be added to guarantee robustness and accuracy. The performance promotion of AEs benefits from both their data-driven property and the physical constraints. However, deep learning-based blind HU is still an ill-posed inverse problem with a large search space. This leads to the unmixing results being sensitive to the hyperparameters of the deep learning model, and the optimal solution cannot be guaranteed. In addition, although AE-based HU introduces physical constraints, its network structures and optimization are in a black-box setting, i.e., there is a lack of interpretability.

In this article, to solve the abovementioned problems, deep unrolling is introduced for blind HU, taking advantage of both data-driven and model-driven methods. Unrolling/unfolding is aimed at designing the network structure based on a conventional iterative algorithm. Specifically, as illustrated in Fig. 1, deep unrolling breaks through the mixing model restrictions of the iterative algorithms, i.e., the NMF-based methods, with a larger search space, due to its data-driven property. Meanwhile, compared with the AE-based methods, deep unrolling-based HU has a faster convergence rate. As a result, deep unrolling has been widely applied to solve various inverse problems in the field of image processing [38], e.g., compressive sensing [39], deconvolution [40], rain removal [41], super-resolution [42], and deblurring [43]. In the field of blind HU, Xiong et al. [44] proposed a novel unrolling-based HU method, termed SNMF-Net, which unrolls a pixelwise sparsity regularized NMF algorithm into a hierarchical deep network architecture and optimizes the objective function in a data-driven manner. Compared with the original iterative algorithm, SNMF-Net is more generalizable and can be applied to more complex scenes. Note that the multilayer deep NMF algorithms, e.g., sparsity-constrained deep NMF with total variation (SDNMF-TV) [14] and self-supervised robust deep matrix factorization (SSRDMF) [45], are also an integration of NMF and deep learning, but they are closer to the AE-based algorithms, with similar encoder–decoder architectures.

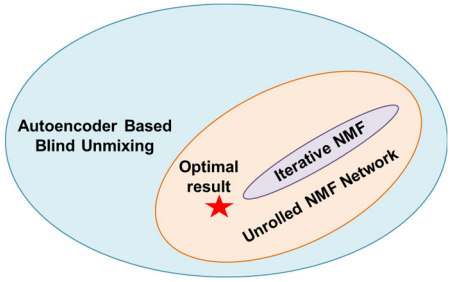


Fig. 1. Illustration of the advantage of unrolling algorithms compared with iterative NMF blind unmixing algorithms and AE-based blind unmixing networks from a functional approximation perspective.

In this article, we propose a group sparsity regularized unmixing unrolling (GSUU) network for blind HU. The GSUU network is a deep neural network, which unrolls a regularized matrix factorization objective function. Specifically, the GSUU network consists of two blocks—the A-Block for endmember estimation and the S-Block for abundance inversion—which work alternately and iteratively to estimate the optimal endmember spectra and abundance maps. The network structures and the learning procedures are designed according to the optimization and update rules of the conventional iterative NMF method, respectively. Therefore, GSUU combines the learnable parameters of deep learning and the mathematical optimization of NMF. In addition, compared with the pixelwise sparsity prior, the spatial group sparsity prior is introduced into GSUU to jointly model the spatial correlation and sparsity of each endmember in the abundance maps. By taking advantage of both model-based and learning-based methods, the GSUU method has high efficiency and interpretability. From the synthetic and real-data experiment results obtained in this study, it is clear that the GSUU method is capable of obtaining more accurate results than the NMF-based and AE-based blind HU algorithms. The main contributions of GSUU can be summarized as follows.

1) A deep unrolled NMF network termed GSUU is proposed for blind HU, which combines the advantages of deep learning and NMF models to iteratively obtain the optimal HU solutions. The GSUU network has a good computational efficiency and performs well for blind HU.
2) A spatial group sparsity prior is used in the S-Block of GSUU to jointly model the spatial correlation and

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sparsity of the abundances, so that the estimated abundances are more in line with the real spatial distributions of the endmembers.

The rest of this article is organized as follows. The problem of blind unmixing is described in Section II. The NMF technique and the unsupervised deep learning-based blind unmixing methods are also introduced. Section III presents the group sparsity mixing model and describes the GSUU method in detail. The experimental analysis conducted with synthetic data is described in Section IV, and that conducted with real hyperspectral data is described in Section V. Finally, our conclusions are drawn in Section VI.

II. Blind Unmixing Model

In this section, the linear mixing model (LMM) is first introduced. NMF-based blind unmixing and unsupervised deep learning for blind HU are then introduced. For ease of understanding, the main notation is summarized in Nomenclature.

A. Linear Mixing Model

Mixing models are aimed at describing the endmember mixing process and can be roughly categorized into LMMs and nonlinear mixing models (NMMs) [46], [47], [48], [49], [50], [51]. The LMM is a simple but effective model for HU that assumes that light only interacts with a single material before being collected by a sensor. Considering its stable and flexible properties, the LMM is widely used in blind HU methods.

An L-band hyperspectral image of N pixels is denoted as $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{L\times N}$. As for the jth pixel of the image $\mathbf{x}_j = [x_{1j}, \ldots, x_{Lj}]^{\mathrm{T}} \in \mathbb{R}^L$, it can be described as follows:

$$\mathbf{x}_j = \sum_{i=1}^{M} s_j[i]\mathbf{a}_i + \boldsymbol{\varepsilon}_j = \mathbf{A}\mathbf{s}_j + \boldsymbol{\varepsilon}_j \tag{1}$$

where $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_M] \in \mathbb{R}^{L\times M}$ is a set of M distinct endmembers, $\mathbf{s}_j = [s_j[1], \ldots, s_j[M]]^{\mathrm{T}} \in \mathbb{R}^M$ represents the abundance vector, and the additive noise is indicated by $\boldsymbol{\varepsilon}_j \in \mathbb{R}^L$. Due to the physical meaning of the abundances, $\mathbf{s}_j$ satisfies two physical constraints, which are called the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC)

$$s_j[i] \geq 0, \quad \sum_{i=1}^{M} s_j[i] = 1. \tag{2}$$

In general, it is assumed that the pixels in a scene share the same set of endmembers $\mathbf{A}$. The LMM can be expressed in matrix notation form as

$$\mathbf{X} = \mathbf{A}\mathbf{S} + \mathbf{N} \tag{3}$$

where $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_N] \in \mathbb{R}^{M\times N}$ is the abundance matrix and $\mathbf{N} = [\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_N] \in \mathbb{R}^{L\times N}$ is the noise matrix.

B. NMF for Blind HU

NMF can obtain two nonnegative components, so based on this, the observed hyperspectral image $\mathbf{X}$ is decomposed into two factor matrices of the endmembers ($\mathbf{A}$) and abundances ($\mathbf{S}$).

As the objective function is nonconvex, some additional constraints are imposed as regularizations of the objective function. Many different types of prior knowledge, such as the geometric property, independence, and sparsity, can be embedded as regularizations. For example, MVCNMF [25] considers the simplex volume of the endmember matrix as the constraint; based on the hyperspectral data characteristics, the blind spectral unmixing method based on sparse component analysis (BSUSCA) [23] takes the abundance sparsity into account for blind unmixing; abundance separation and smoothness constrained NMF (ASSNMF) [52] adds an abundance separation constraint by adopting the Kullback–Leibler (K-L) divergence and an abundance smoothness constraint by computing the abundance differences; and $L_{1/2}$NMF [27] extends NMF by using the $L_{1/2}$ sparsity constraint of the abundance.

In general, an abundance regularizer considering the sparsity can increase the uniqueness of the results. The optimization problem considering sparsity regularization can be categorized as follows:

$$\min_{\mathbf{A}\geq 0, \mathbf{S}\geq 0} f(\mathbf{A}, \mathbf{S}) = \|\mathbf{X} - \mathbf{A}\mathbf{S}\|_{\mathrm{F}} + \lambda g(\mathbf{S}) \text{ s.t. } \mathbf{A} \geq 0, \ \mathbf{S} \geq 0 \tag{4}$$

in which $\|\cdot\|_{\mathrm{F}}$ represents the Frobenius norm, $\geq$ permits the matrix to be nonnegative, λ is the parameter of the regularizer used to control the contribution of the data fitting and the sparsity regularization, and regularization $g(\cdot)$ represents the sparsity property of the abundance $\mathbf{S}$. Traditionally, the commonly used sparseness measure is the $\ell 0$-norm [53], [54], but the abundance measure function $g(\mathbf{S})$ is not differentiable so that the $\ell 0$-norm is often replaced by the $\ell 1$-norm, which is a widely used sparsity regularizer [55].

In recent years, mixed-norm regularizers considering sparsity have been used in many fields [56], [57]. In NMF-based HU, the collaborative NMF (CoNMF) algorithm, which assumes that the estimated number of endmembers $\hat{M} > M$ and the true number of endmembers, uses a collaborative regularization prior to promote the row sparsity of $\mathbf{S}$, which can be described as $g(\mathbf{S}) = \sum_{i=1}^{\hat{M}} \left\|s^i\right\|_2^q$, where $0 < q \leq 1$ is used to control the degree of collaboration.

C. Unrolling for NMF

Deep learning is a "black box" as it lacks interpretability. Deep unrolling involves designing the deep network architecture by mapping the model-based method of each iteration to a typical layer of a deep neural network and fixing a limited number of layers. Benefiting from both model-based learning and deep learning, unrolled networks naturally inherit the respective advantages of both approaches [38]. Algorithm unrolling, or unfolding, increases the interpretability by connecting an iterative algorithm to a neural network architecture, where one layer of the network corresponds to one algorithm iteration step [58]. When the network is passed through, it is equal to conducting the iterative algorithm, and the parameters of the iterative algorithm are replaced by the

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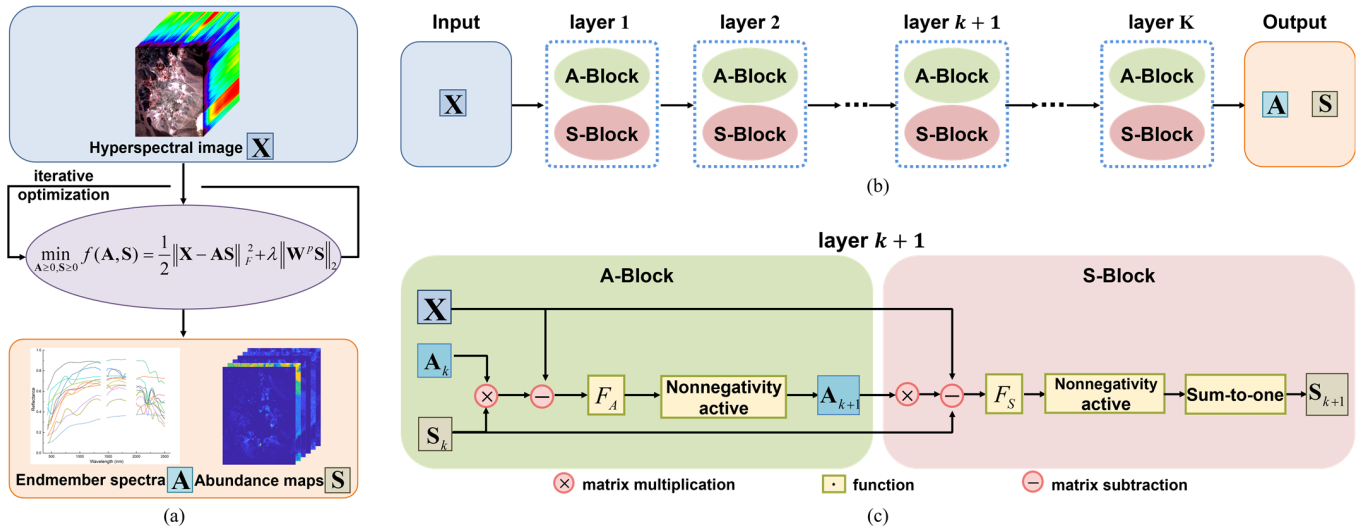


Fig. 2. Flowchart of the proposed algorithm: (a) procedure of the spatial group sparsity constrained NMF and optimized conventional iterative algorithm for estimating the endmember spectra and abundance maps; (b) structure of the GSUU network; and (c) processing details of the $k+1$th layer of the A-Block and S-Block.

network parameters. Unrolling methods were first developed to solve the sparse coding problem by proposing fast neural network approximations [38], and unrolling has also been used to solve many signal and image processing problems [39], [40], [41], [42], [43]. Compared with the advanced deep learning networks, there are generally fewer parameters in the conventional iterative algorithms. Furthermore, the unrolled architecture of the network is knowledge-oriented and is designed based on the prior knowledge of the domain.

Blind unmixing is used for solving HU problems by conducting inversion of the endmember spectra and abundances at the same time. NMF is a significant category of blind unmixing as it can meet the need for the endmember spectra and abundances to be nonnegative. The basic NMF algorithms are sensitive to the initialization, so considering the prior knowledge can better solve this problem. Based on this, many constrained NMF algorithms for blind unmixing have been proposed [23], [24], [25], [26], [27], [28]. In effect, the iterative algorithms are used to solve the nonconvex problem.

Therefore, deep unrolling can be used for blind unmixing by unrolling the optimization problem into the network architecture, which is designed based on the update rules of the iterative algorithm. The update of the algorithm steps is then replaced by the network layers. Some HU networks based on deep unrolling have been proposed in recent years. For example, based on the LMM and the iterative shrinkage-thresholding algorithm (ISTA), the spectral mixture model-inspired neural network for blind unmixing (MNN-BU) developed by Qian et al. [59] conducts inversion of the abundance maps based on unrolling. In addition, the SNMF-Net method based on the LMM, as proposed by Xiong et al. [44], is an interpretable unmixing network based on unrolling the optimization of L_p sparsity constrained nonnegative matrix factorization (L_p-NMF).

III. GSUU Algorithm

A detailed description of the GSUU algorithm is presented in this section. The basic framework of the GSUU algorithm is illustrated in Fig. 2. First, the group sparsity mixing model is established, as shown in Fig. 2(a), inspired by the group sparsity of the distribution of the materials. This is then unrolled to the GSUU network containing two subnets of each layer, called the A-Block and the S-Block, as shown in Fig. 2(b). The two blocks are, respectively, used to update the endmembers and abundances in every layer. The detailed structures of the two blocks are illustrated in Fig. 2(c).

A. Spatial Group Sparsity Prior to the Abundances

Since ground objects are often sparsely distributed, sparsity is a vital prior to the abundances, as shown in Fig. 3. NMF-based HU considering the sparsity property has been widely studied.

As shown in Fig. 3, the distribution of the ground objects is sparse in the local regions of the hyperspectral image, which means that the objects within a group are more likely to share the same sparse distribution. Therefore, with the spatial neighbors having spectral similarities, the hyperspectral image can be segmented into predefined spatial groups (superpixels) $\{\vartheta_1, \ldots, \vartheta_P\}$, where each group ϑ shares the same sparse pattern. A new group sparsity regularizer $g(\mathbf{S}) = \sum_{p=1}^{P} \sum_{s_j \in \vartheta_p} \|\mathbf{W}^p \mathbf{s}_j\|_2$ is used in the NMF-based blind unmixing to better describe the sparse representation in each group. Instead of exploring the spatial information of the whole image, the sparsity property is explored at the spatial group level.

In this study, in order to generate superpixels, we used a representative gradient ascent-based algorithm called SLIC [60]. This algorithm is simple in structure and fast, and there are only two parameters, P and W_S, that need to be adjusted. P is used to decide the ideal number of superpixels, and W_S is intended to balance the spectral and spatial similarities.

Based on the SLIC algorithm, the pixels in the hyperspectral image $\mathbf{X}$ are categorized into spatial groups denoted with a matrix $\mathbf{X}_s = (\mathbf{X}^1, \ldots, \mathbf{X}^P) \in \mathbb{R}^{L \times N}$, in which P denotes the superpixel number. Correspondingly, the abundance matrix

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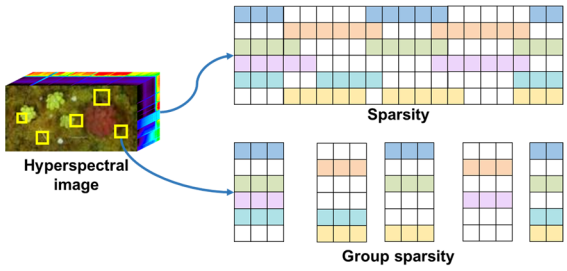


Fig. 3. Sparsity and group sparsity are shown in the image.

is categorized into P groups, which indicates that $S_s = (S^1, \ldots, S^P) \in \mathbb{R}^{M\times N}$. Therefore, the abundance matrix of the pixels in the same superpixel should have the same nonzero structure since the materials in the same superpixel are the same. In addition, since only a few endmembers are in a continuous region, the structure of the abundance in the same superpixel should be sparse.

In order to promote the spatial group sparsity, a spatial group sparsity regularizer is taken into account for the abundance, so that the objective function is described as follows:

$$\min_{\mathbf{A}\geq 0,\mathbf{S}\geq 0} f(\mathbf{A},\mathbf{S}) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}\mathbf{S}\|_{\mathrm{F}}^2 + \lambda\left\|\mathbf{W}^p\mathbf{S}\right\|_2 \tag{5}$$

where $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_N] \in \mathbb{R}^{M\times N}$ denotes the abundance matrix, and the regularization parameter λ balances the reconstruction of the data and the group sparsity regularizer. $\mathbf{W}^p = [\mathbf{w}_1^p, \ldots, \mathbf{w}_N^p]^{\mathrm{T}} \in \mathbb{R}^{M\times N}$ denotes the superpixelwise weight. The pth ($p = 1, 2, \ldots, P$) superpixel weights can be expressed as follows:

$$\bar{\mathbf{s}}^p \leftarrow \arg\min \frac{1}{2}\|\bar{\mathbf{x}}^p - \mathbf{A}\bar{\mathbf{s}}^p\|_{\mathrm{F}}^2 \ \text{ s.t. } \bar{\mathbf{s}}^p \geq 0, \quad \mathbf{1}^{\mathrm{T}}\bar{\mathbf{s}}^p = 1 \tag{6}$$

$$\mathbf{w}_i^p = \frac{1}{|\bar{s}^p[i]| + \varepsilon} \tag{7}$$

where $\bar{\mathbf{x}}^p$ is an average spectral vector of the pixels in the pth superpixel, $\bar{\mathbf{s}}^p = [\bar{\mathbf{s}}^p[1], \ldots, \bar{\mathbf{s}}^p[M]]^{\mathrm{T}} \in \mathbb{R}^{M\times 1}$ represents the average abundance vector of the pth superpixel, and the positive parameter ε is used to prevent infinite weights. In addition, when the endmember matrix $\mathbf{A}$ is updated, the weight matrix $\mathbf{W}^p$ will update as well in every iteration.

For the proposed objective function considering group sparsity (5), the conventional optimization algorithms consist of updating the three matrices $\mathbf{W}^p$, $\mathbf{S}$, and $\mathbf{A}$. $\mathbf{W}^p$ can be calculated by (6) and (7). Therefore, the main parts are the updating of $\mathbf{S}$ and $\mathbf{A}$. We use the alternating least squares (ALS) algorithm [61] to optimize this problem so that the objective function for the endmember matrix $\mathbf{A}$ and abundance matrix $\mathbf{S}$ corresponds to

$$\mathbf{A}_{k+1} \leftarrow \arg\min_{\mathbf{A}\geq 0} f(\mathbf{A},\mathbf{S}_k) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}\mathbf{S}_k\|_{\mathrm{F}}^2 \tag{8}$$

$$\mathbf{S}_{k+1} \leftarrow \arg\min_{\mathbf{S}\geq 0,\mathbf{1}^{\mathrm{T}}\mathbf{S}=1} f(\mathbf{A}^{k+1},\mathbf{S}) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}_{k+1}\mathbf{S}_k\|_{\mathrm{F}}^2 + \lambda\left\|\mathbf{W}^p\mathbf{S}\right\|_2 \tag{9}$$

where k is the iteration number.

Based on the nonnegativity constrains of endmembers and abundance, the endmember matrix $\mathbf{A}$ and abundance matrix $\mathbf{S}$ can be calculated by iterative gradient descent, and the update rule corresponds to

$$\mathbf{A}_{k+1} = \max\left(0, \mathbf{A}_k - \omega_k(\mathbf{A}_k\mathbf{S}_k - \mathbf{X})\mathbf{S}_k^{\mathrm{T}}\right) \tag{10}$$

$$\mathbf{S}_{k+1} = \max(0, \mathbf{S}_k - \sigma_k(\mathbf{A}_{k+1}^{\mathrm{T}}(\mathbf{A}_{k+1}\mathbf{S}_k - \mathbf{X}) + \lambda\mathbf{W}^{p\mathrm{T}})) \tag{11}$$

where ω_k and σ_k are the step size. $\mathbf{W}^p$ represents the pixelwise weight matrix.

In general, the update rules of conventional iterative algorithms, which contain analytic operations, e.g., matrix multiplication and matrix subtraction, are of the same characters as a network, and thus, the update rules (10) and (11) can be explicitly modeled as a network layer as shown in Fig. 2(c), while the K times iteration can be considered as a deep network with K layers cascaded. The details will be illustrated in Section III-B.

B. Unrolling of Group Sparsity Regularized Unmixing

According to the optimization of the objective function, the network contains two blocks, called the A-Block and the S-Block, for iteratively updating the endmember spectra and abundances, as shown in Fig. 2(b). The details of the two blocks are shown in Fig. 2(c).

1) A-Block for Endmember Extraction: Based on the iterative update rules of the endmember matrix $\mathbf{A}$ in (10), the $k+1$th layer of the subnetwork for updating the endmembers $\mathbf{A}$ can be defined as follows:

$$\mathbf{A}_{k+1} = \eta(\mathbf{A}_k - \rho_{1k}(\mathbf{A}_k\mathbf{S}_k - \mathbf{X})\mathbf{W}_{1k}) \tag{12}$$

where the nonnegativity of $\mathbf{A}$ is satisfied by the use of the rectified linear unit (ReLU) activation function $\eta(\cdot)$, where ρ_{1k} is the step size and $\mathbf{W}_{1k}$ is the network parameter. The total procedure for the network parameters is denoted as F_A in Fig. 2(c).

It can be found that the size of $\mathbf{W}_{1k}$ and the input data is the same, which means that, with the limitation of the matrix multiplication rules, the limitation of the network is such that the network changes with different sizes of input image. In fact, any number of pixels can be used to test the network. Moreover, the output results of the A-Block produce a collection of endmembers based on the input pixels, and when a different set of pixels is given, the output results may change. As for the fact that there is only a single collection of endmembers in a scene, the pixels in the same data should have the same endmembers. Therefore, in order to satisfy this, the network weight $\mathbf{W}_{1k}$ is replaced by the abundance matrix $\mathbf{S}_k^{\mathrm{T}}$, which allows the network to input any size of the image. Accordingly, the endmember update based on the A-Block follows the iteration as follows:

$$\mathbf{A}_{k+1} = \eta\left(\mathbf{A}_k - \rho_{1k}(\mathbf{A}_k\mathbf{S}_k - \mathbf{X})\mathbf{S}_k^{\mathrm{T}}\right). \tag{13}$$

2) S-Block for Abundance Inversion: In order to meet the physical interpretation of the abundances, every column of $\mathbf{S}_{k+1}$ should satisfy the sum-to-one constraint (ASC)

$$\mathbf{s}_i \leftarrow \frac{\mathbf{s}_i}{\sum_i^M \mathbf{s}_i}. \tag{14}$$

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Differing from the updating of the endmember matrix $\mathbf{A}$, the abundance matrix $\mathbf{S}$ needs additional physical constraints, i.e., the ANC and the ASC. Therefore, the S-Block for abundance $\mathbf{S}$ can be designed by repetition. Consequently, the $k+1$th layer in the S-Block is expressed as

$$\mathbf{S}_{k+1} = \xi(\eta(\mathbf{S}_k - \rho_{2_k}(\mathbf{W}_{2_k}(\mathbf{A}_{k+1}\mathbf{S}_k - \mathbf{X}) + \lambda \mathbf{W}^{p\mathrm{T}}))) \quad (15)$$

where $\xi(\cdot)$ is a normalization layer to satisfy the ASC in (14). In order to satisfy the nonnegativity of the abundances $\mathbf{S}$, the ReLU activation function $\eta(\cdot)$ is used, ρ_{2k} is the step size, and $\mathbf{W}_{2k}$ is the network parameter. The total procedure for the network parameters is denoted as F_{S} in Fig. 2(c).

3) Implementation of the Network: In this study, the mean squared error (MSE) computing the Euclidean distance between the input and reconstructed data was chosen as the loss function, which is expressed as follows:

$$L = \frac{1}{2N}\sum_{k=1}^{K} \|\mathbf{X} - \mathbf{A}_k\mathbf{S}_k\|_{\mathrm{F}}^2. \quad (16)$$

The endmember matrix $\mathbf{A}_0$ is estimated using any endmember extraction algorithm, e.g., VCA. The initial abundances $\mathbf{S}_0$ can be calculated by the use of any traditional unmixing method, such as FCLS.

IV. Synthetic Experiments

An experimental analysis was conducted between the proposed algorithm and the following advanced blind unmixing methods—NFINDR-FCLS [12], [18], $\mathrm{L}_{1/2}$NMF [27], SGSNMF [28], uDAS [32], CyCU-Net [36], UnDIP [62], and SNMF-Net [44]—using synthetic data simulated with different mixing models. The main purpose was to assess the performance of the proposed algorithm in both endmember extraction and abundance inversion.

A. Performance Evaluation Indices

In order to analyze the endmember extraction accuracy, the SAD is introduced

$$\mathrm{SAD}_i = \arccos\frac{\hat{\mathbf{a}}_i^{\mathrm{T}}\mathbf{a}_i}{\|\hat{\mathbf{a}}_i\|_2\|\mathbf{a}_i\|_2} \quad (17)$$

where $\mathbf{a}_i$ and $\hat{\mathbf{a}}_i$ are the ground-truth endmember spectra and the estimated spectra of the ith endmember, respectively. Note that the unit for this metric is degree. The abundance estimation accuracy is assessed using the root-mean-square error (RMSE), which can be expressed as follows:

$$\mathrm{RMSE}_i = \frac{1}{\sqrt{N}}\|\mathbf{S}_i - \hat{\mathbf{S}}_i\|_{\mathrm{F}} \quad (18)$$

where $\mathbf{S}_i$ is the estimated abundance, $\hat{\mathbf{S}}_i$ is the corresponding abundance ground truth, and N is the number of pixels.

B. Synthetic Data

A synthetic dataset—Synth-1—was used for the evaluation. This simulated data cube, as shown in Fig. 4, was generated using five randomly selected spectral signatures from the United States Geological Survey (USGS) library. Data cube

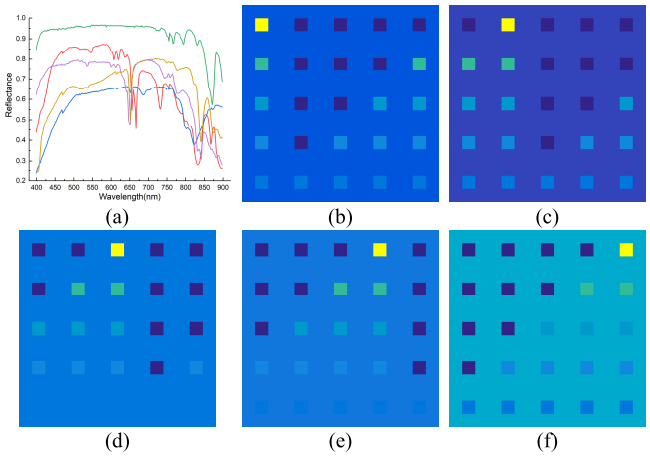


Fig. 4. Synth-1 data: (a) five endmember spectra and (b)–(f) abundances maps for the five endmembers.

#1 (DC1) has 75 × 75 pixels and 224 bands per pixel [63]. The data were generated based on an LMM and a generalized bilinear mixing (GBM) model, with five endmembers, and the abundances met the ASC in each simulated pixel. In the resulting simulated image, there are both pure regions and mixed regions, with the number of endmembers ranging from two to five, with a square block distribution. In addition, the background pixels are mixed with the same five endmembers.

In order to simulate noisy hyperspectral images, white Gaussian noise was added. The signal-to-noise ratio (SNR) of the data was defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{E\left[\mathbf{x}^{\mathrm{T}}\mathbf{x}\right]}{E\left[\boldsymbol{\varepsilon}^{\mathrm{T}}\boldsymbol{\varepsilon}\right]} \quad (19)$$

where $\mathbf{x}$ is the clear hyperspectral image, and ε is the noise.

1) Influence of the Number of Layers: How the number of layers K influences the unmixing accuracy was first analyzed. The clean hyperspectral image mixed by the LMM was degraded by Gaussian noise with SNR = 20 and 30 dB, and the number of epochs was set to 300. By altering K, we recorded the changes of SAD and RMSE, with K ranging from 3 to 30 with a step size of 2, as shown in Fig. 5(a) and (b), respectively. As the results show, the mean SAD values for the GSUU method under different noise levels do not change much with the increasing number of layers; however, the values for SNMF-Net generally increase with the number of layers. The values for GSUU are smaller than those for SNMF-Net with the same number of layers. As for the RMSE, this shows similar results. In the subsequent experiments, we set the number of layers to nine, which is the same as SNMF-Net.

2) Influence of Different Mixing Models: The goal of this experiment was to analyze the sensitivity of different mixing models (LMM and GBM) with noise levels of 20 and 30 dB. Tables I and II, respectively, list the mean SAD and RMSE values, plus their standard deviation, under different mixing models with 20- and 30-dB noise levels. The results consist of the mean and standard deviation of ten times random tests, and the best result at each SNR level is highlighted in red. It can be seen that, under the LMM, the deep learning-based methods

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TABLE I

PERFORMANCE OF ALL EIGHT UNMIXING METHODS WITH RESPECT TO DIFFERENT MIXING MODELS AND NOISE LEVELS, IN TERMS OF THE SAD, WITH THE HIGHEST ACCURACIES HIGHLIGHTED IN RED

Mixing model	*SNR (dB)*	*NFINDR-FCLS*	*$L_{1/2}$NMF*	*SGSNMF*	*uDAS*	*CyCU-Net*	*UnDIP*	*SNMF-Net*	*GSUU*
LMM	*20*	5.87±0.09	1.60±0.37	2.99±1.44	1.89±0.06	1.87±0.14	1.77±0.09	1.54±0.08	**1.51±0.06**
	30	1.89±0.02	0.60±0.44	0.41±0.06	0.78±0.40	1.26±0.35	0.41±0.02	0.41±0.11	**0.38±0.05**
GBM	*20*	6.93±0.48	3.05±0.40	4.06±2.02	3.28±0.71	2.52±0.44	2.73±0.44	2.06±0.23	**1.97±0.28**
	30	2.88±0.10	2.49±0.52	3.32±0.14	4.00±0.55	1.94±0.33	2.00±0.04	**1.32±0.15**	1.42±0.23

TABLE II

PERFORMANCE OF ALL EIGHT UNMIXING METHODS WITH RESPECT TO DIFFERENT MIXING MODELS AND NOISE LEVELS, IN TERMS OF THE RMSE, WITH THE HIGHEST ACCURACIES HIGHLIGHTED IN RED

Mixing model	*SNR (dB)*	*NFINDR-FCLS*	*$L_{1/2}$NMF*	*SGSNMF*	*uDAS*	*CyCU-Net*	*UnDIP*	*SNMF-Net*	*GSUU*
LMM	*20*	0.16±0.01	0.17±0.08	0.16±0.04	0.12±0.01	0.57±0.08	0.38±0.01	0.22±0.00	**0.11±0.01**
	30	0.05±0.00	0.07±0.08	0.06±0.02	0.06±0.00	0.44±0.12	0.40±0.00	0.19±0.01	**0.04±0.01**
GBM	*20*	0.24±0.08	0.42±0.07	0.64±0.10	0.40±0.06	0.54±0.05	0.42±0.02	0.26±0.03	**0.17±0.05**
	30	0.34±0.05	0.38±0.05	0.47±0.01	0.31±0.04	0.46±0.07	0.42±0.00	0.21±0.00	**0.14±0.04**

TABLE III

PERFORMANCE OF ALL EIGHT UNMIXING METHODS WITH RESPECT TO DIFFERENT NOISE LEVELS, IN TERMS OF THE SAD, WITH THE HIGHEST ACCURACIES HIGHLIGHTED IN RED

SNR (dB)	*NFINDR-FCLS*	*$L_{1/2}$NMF*	*SGSNMF*	*uDAS*	*CyCU-Net*	*UnDIP*	*SNMF-Net*	*GSUU*
20	5.87±0.09	1.60±0.37	2.99±1.44	1.89±0.06	1.87±0.14	1.77±0.09	1.54±0.08	**1.51±0.06**
30	1.89±0.02	0.60±0.44	0.41±0.06	0.78±0.04	1.26±0.35	0.41±0.02	0.41±0.11	**0.38±0.05**
40	0.60±0.01	0.11±0.01	0.79±0.08	0.80±0.01	0.17±0.00	0.11±0.01	0.11±0.03	**0.10±0.01**
50	0.19±0.00	0.03±0.00	0.93±0.05	0.81±0.00	0.13±0.00	0.04±0.00	**0.03±0.00**	**0.03±0.00**

TABLE IV

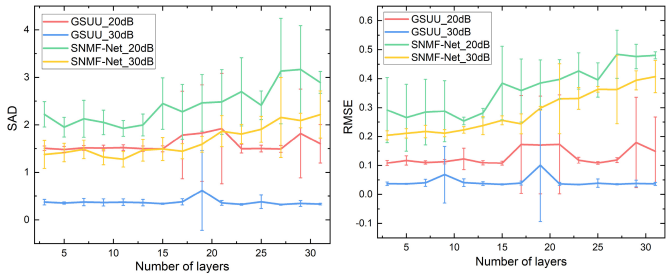
PERFORMANCE OF ALL EIGHT UNMIXING METHODS WITH RESPECT TO DIFFERENT NOISE LEVELS, IN TERMS OF THE RMSE, WITH THE HIGHEST ACCURACIES HIGHLIGHTED IN RED

SNR (dB)	*NFINDR-FCLS*	*$L_{1/2}$NMF*	*SGSNMF*	*uDAS*	*CyCU-Net*	*UnDIP*	*SNMF-Net*	*GSUU*
20	0.16±0.01	0.12±0.01	0.16±0.04	0.12±0.01	0.57±0.08	0.38±0.01	0.22±0.00	**0.11±0.01**
30	0.05±0.00	0.04±0.00	0.06±0.02	0.06±0.00	0.44±0.12	0.40±0.00	0.19±0.01	**0.04±0.01**
40	**0.01±0.00**	0.42±0.07	0.16±0.01	0.04±0.00	0.55±0.03	0.40±0.00	0.18±0.00	**0.01±0.00**
50	**0.00±0.00**	0.38±0.05	0.18±0.01	0.04±0.00	0.62±0.07	0.40±0.00	0.18±0.00	**0.00±0.00**

perform better than the traditional methods for endmember extraction, and the GSUU method performs better than the other algorithms. As for the endmember extraction for the GBM data, the GSUU method performs the best under the case of a 20-dB noise level, whereas SNMF-Net performs the best under the case of a 30-dB noise level. From Table II, it is apparent that the GSUU method has the advantage in abundance inversion as it considers the spatial information of the hyperspectral imagery.

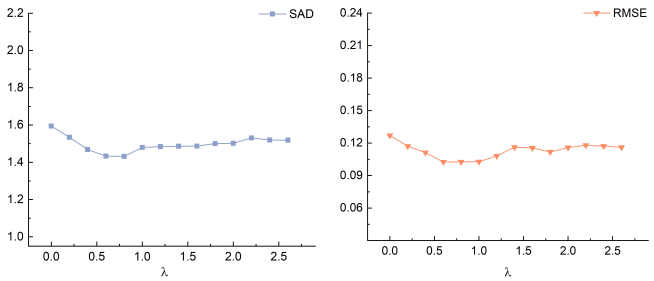
3) Robustness to Noise: This experiment was conducted to assess the robustness of the algorithms. Synthetic data mixed by the LMM were polluted by white Gaussian noise with noise levels ranging from 20 to 50 dB, with an interval of 10 dB. The results obtained with the four noise levels are listed in Tables III and IV, where the mean SAD value is listed with its standard deviation of ten times measurements, and the best result for each SNR level is highlighted in red. From Table III, it is apparent that the GSUU method extracts the best endmember spectra results with the more noisy mixed data, compared with the other methods. The GSUU method can also obtain more accurate results for the less noisy mixed data. From Table IV, it is apparent that the GSUU method can obtain more accurate abundance maps than the other methods.

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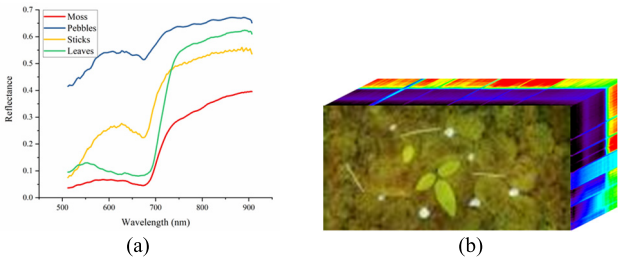
(a) (b)

Fig. 5. Performance of GSUU and SNMF-Net with different numbers of layers and different noise levels, in terms of the: (a) SAD and (b) RMSE.



(a) (b)

Fig. 6. Performance analysis of GSUU with respect to λ: (a) SAD and (b) RMSE.



(a) (b)

Fig. 7. RMMS dataset: (a) endmember spectra and (b) scene 1 mixed data cube.

4) Parameter Sensitivity Analysis: Parameter λ was analyzed with synthetic data mixed by LMM in the condition of 20-dB white Gaussian noise. Fig. 6 illustrates the tendency of the unmixing performance evaluated by SAD and RMSE as λ increases from [0, 2.6]. As shown in Fig. 6, better performance is obtained when the parameter λ changing from 0 to 0.8, which efficiently shows the usefulness of the group sparsity regularization. When the parameter λ is larger than 1.0, the performance is worse for balancing the reconstruction and the group sparsity, but the performance is stable and better than no group sparsity regularization one.

V. Real-Data Experiments

In this section, we describe how the analysis of the performance of different algorithms was conducted based on two real datasets captured by airborne and ground-borne hyperspectral sensors. The results are the mean of ten times random tests.

A. RMMS Dataset

The realistic mixing miniature scenes (RMMS) [64] dataset[1] contains two scenes. SMScene made up of homogeneous flat materials, as shown in Fig. 7, is used for analyzing the performance. SMScene is collected by multiscale observation using an imec SNAPSCAN visible and near-infrared (VNIR) system. Therefore, the ground-truth abundance is available for quantitative evaluation of HU results. It contains the point, line, and area characteristics, and the spectral similarity of the materials increases the challenge of spectral unmixing. The spatial resolution of the unmixing image is 2 mm/pixel. The image size is 64 × 128 pixels with 132 spectral channels after noisy band removal. SMScene contains four endmembers, which are named moss, pebbles, sticks, and leaves.

For SMScene, the experimental results of the endmember extraction and abundance inversion are, respectively, listed in Tables V and VI. Fig. 8 shows the abundances maps for all the compared methods. As can be seen in Table V, the proposed GSUU method can obtain the state-of-the-art results in average SAD values. The GSUU method can also estimate the abundance maps more accurately than the other methods. For the sticks endmember, GSUU performs better

TABLE V

Endmember Extraction for SMScene of the RMMS Dataset, With the Highest Accuracies Highlighted in Red

Method	*Endmember*				
	Moss	*Pebbles*	*Sticks*	*Leaves*	*Average*
NFINDR-FCLS	15.4388	2.1372	3.5705	**0.9015**	5.5100
$L_{1/2}$NMF	17.9585	2.5511	3.9829	2.0965	6.6472
SGSNMF	2.3242	2.5392	19.8028	1.0235	6.4224
uDAS	19.2617	2.8042	5.5605	2.5720	7.5496
CyCU-Net	**2.2060**	5.4470	10.0200	4.6425	5.5789
UnDIP	14.2494	**2.1487**	3.6208	1.2293	5.3120
SNMF-Net	13.3987	2.3738	**3.3066**	2.3085	5.3469
GSUU	7.2930	2.3337	3.3854	1.0964	**3.5271**

TABLE VI

Abundance Inversion for SMScene of the RMMS Dataset, With the Highest Accuracies Highlighted in Red

Method	*Endmember*				
	Moss	*Pebbles*	*Sticks*	*Leaves*	*RMSE*
NFINDR-FCLS	0.4516	0.0412	0.0839	0.3870	0.6020
$L_{1/2}$NMF	0.4573	0.0423	0.0884	0.3865	0.6068
SGSNMF	0.5768	0.0551	0.4003	**0.2157**	0.7365
uDAS	0.4617	0.0466	0.0855	0.3837	0.6081
CyCU-Net	0.4616	0.0665	0.2012	0.3493	0.6369
UnDIP	0.7599	0.3612	0.3591	0.3969	0.9972
SNMF-Net	0.4196	0.0448	0.0817	0.3488	0.5538
GSUU	**0.3610**	**0.0354**	**0.0548**	0.3291	**0.4930**

[1]https://doi.org/10.17632/6bg6342xg9.1

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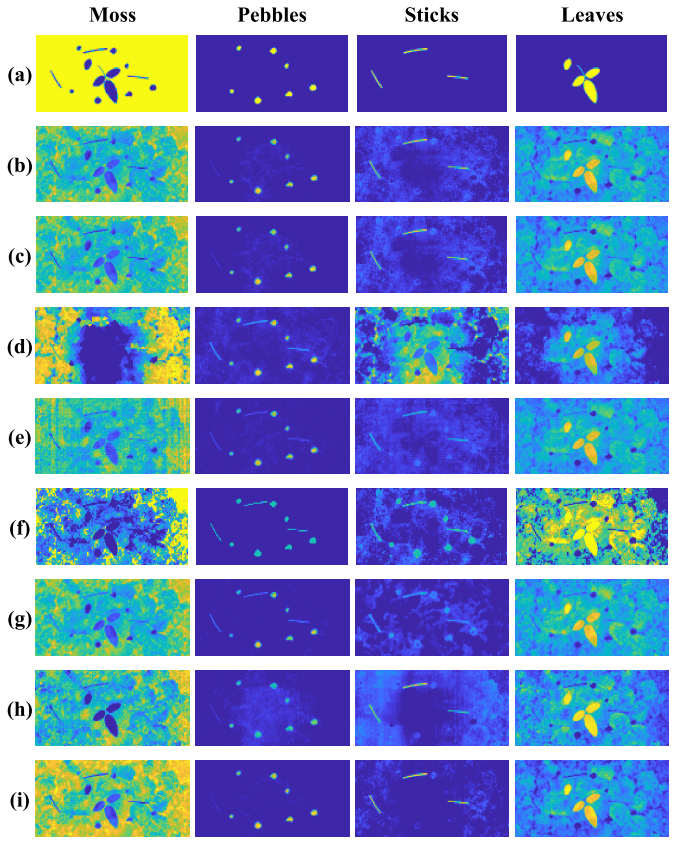


Fig. 8. Comparison of the abundance maps for SMScene of the RMMS dataset: (a) ground truth; (b) NFINDR-FCLS; (c) $L_{1/2}$NMF; (d) SGSNMF; (e) uDAS; (f) CyCU-Net; (g) UnDIP; (h) SNMF-Net; and (i) GSUU.

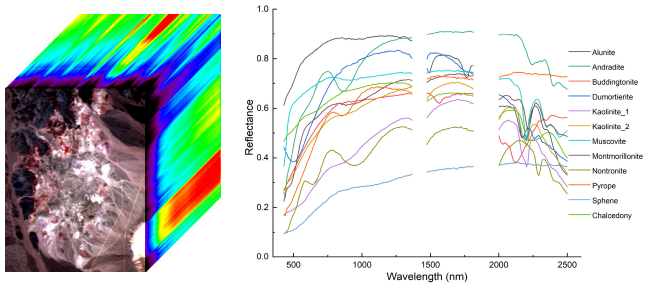


Fig. 9. Cuprite hyperspectral data: (a) hyperspectral image data cube and (b) 12 reference endmember spectra.

than SGSNMF, which also takes into account the group sparsity of the hyperspectral imagery. The SAD accuracies for the pebbles and sticks endmembers separately extracted by UnDIP and SNMF-Net are the best, and neither of them takes into account the spatial information. The abundance map for the sticks endmember of SGSNMF is not reasonable, as the sticks are mistakenly inversed as moss.

B. AVIRIS Cuprite Dataset

The Cuprite dataset with a 0.10-μm spectral resolution is a widely used benchmark dataset for HU [65], which was collected by NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the Cuprite mining site in NV, USA. In this experiment, the selected data containing

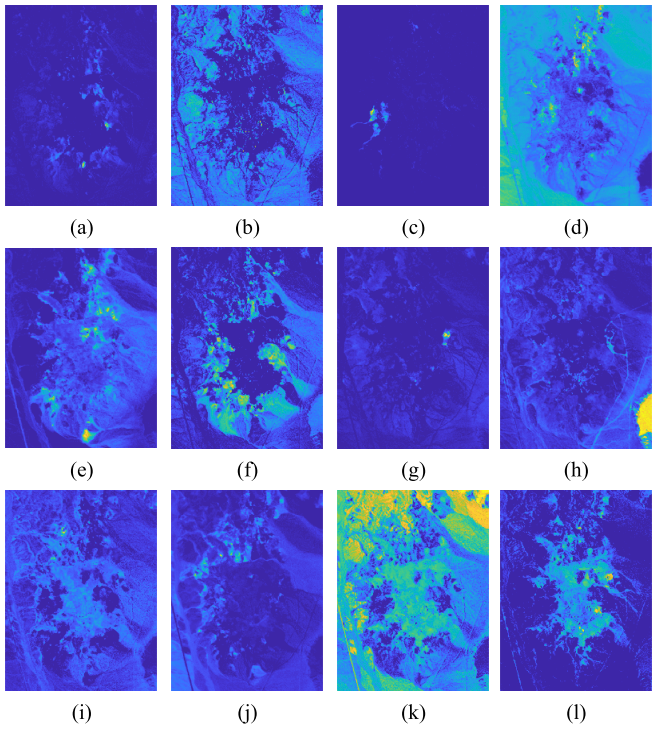


Fig. 10. Abundance maps obtained by the proposed GSUU method: (a) alunite; (b) andradite; (c) buddingtonite; (d) dumortierite; (e) kaolinite_1; (f) kaolinite_2; (g) muscovite; (h) montmorillonite; (i) nontronite; (j) pyrope; (k) sphene; and (l) chalcedony.

250 $\times$ 190 pixels were a subset of the Cuprite dataset, as shown in Fig. 9(a). The subset contained 188 bands as the low-SNR bands were removed. In order to evaluate the performance of the endmember extraction results, 12 mineral spectra, as shown in Fig. 9(b), were chosen from the USGS library as the ground-truth references [21], [28]. The 12 endmembers were alunite, andradite, buddingtonite, dumortierite, kaolinite_1, kaolinite_2, muscovite, montmorillonite, nontronite, pyrope, sphene, and chalcedony.

The SAD values are listed in Table VII, where the best results are highlighted in red. From Table VII, it is apparent that the proposed GSUU method obtains the best average SAD results. Although the abundances cannot be quantitatively evaluated, the abundance maps obtained by the GSUU method are shown in Fig. 10 and can be visually interpreted. Based on the geological maps of the Cuprite area [66], the abundance maps correlate well, and the buddingtonite, muscovite, and montmorillonite, which have distinct distributions, are inversed well by the proposed GSUU method.

C. Efficiency Analysis With Real Hyperspectral Data

Two real hyperspectral datasets were used to verify the efficiency of the proposed method. The running times of different methods are listed in Table VIII. All the algorithms were compared on a personal computer equipped with an Intel Core CPU at 3.60 GHz, 16-GB RAM, and an NVIDIA GTX 1070 GPU. The results are the average of ten runs. It can be seen from Table VIII that the conventional algorithms based on MATLAB require less time. As for the Python-based algorithms, the efficiency of GSUU is high when compared with

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TABLE VII
ENDMEMBER EXTRACTION ACCURACY FOR THE CUPRITE DATASET, WITH THE HIGHEST ACCURACIES HIGHLIGHTED IN RED

Endmember	*Method*							
	NFINDR	*$L_{1/2}$NMF*	*SGSNMF*	*uDAS*	*CyCU-Net*	*UnDIP*	*SNMF-Net*	*GSUU*
Alunite	5.6494	5.5176	6.2624	7.1687	5.3845	**4.5377**	5.3760	5.2476
Andradite	4.9330	4.9561	**3.8789**	6.7882	5.2430	4.7931	4.9077	4.3064
Buddingtonite	**4.3128**	4.6295	4.5206	5.3812	5.2820	5.7141	5.5641	5.3462
Dumortierite	**4.1393**	4.8530	4.8873	6.0983	4.7126	7.8535	6.0077	4.9481
Kaolinite_1	4.7133	4.2857	**4.1138**	5.2885	5.4953	6.4554	4.8986	4.7731
Kaolinite_2	4.3170	3.8102	4.0394	4.5533	4.0528	4.1973	3.7883	**3.6285**
Muscovite	8.5759	3.5008	**3.3174**	8.5364	8.5397	5.7324	7.2903	6.2349
Montmorillonite	3.3871	7.2536	7.5057	4.1636	3.3312	3.4641	3.4021	**3.1225**
Nontronite	**3.7317**	4.5722	4.4003	5.5073	3.9914	3.7337	4.0771	4.1881
Pyrope	**4.1657**	6.7552	5.0649	13.7326	11.2673	18.6232	9.9276	8.5989
Sphene	16.3486	11.4706	8.1818	6.1130	5.4392	22.0718	**5.1292**	5.4607
Chalcedony	4.9272	4.4404	6.6349	6.3502	5.2407	4.7256	**3.9020**	3.9339
Average	5.7667	5.5037	5.2340	6.6401	5.6650	7.6585	5.3559	**4.9824**

TABLE VIII
EFFICIENCY COMPARISON FOR THE REAL HYPERSPECTRAL DATA

Platform	*Method*	*Running time (s)*	
		RMMS	*Cuprite*
MATLAB	*NFINDR-FCLS*	2.16	**64.59**
	$L_{1/2}$NMF	**2.08**	169.19
	SGSNMF	5.27	72.91
	uDAS	98.42	825.27
Python	*CyCU-Net*	90.45	302.53
	UnDIP	53.85	390.06
	SNMF-Net	99.35	943.83
	GSUU	**51.86**	**158.51**

that of the other methods. We believe that the computational cost of GSUU is an acceptable balance of accuracy and time.

VI. CONCLUSION

In this article, a model-driven deep neural network named the GSUU network has been proposed for blind unmixing, to improve the efficiency and performance of HU. Inspired by the fact that spatially adjacent mixed pixels share similar sparse abundances, the spatial group sparsity prior to the abundances is added into the deep unrolling network. Based on the optimization procedure, the network contains two blocks: the A-Block and the S-Block. The A-Block is used to obtain the endmember spectra, and the S-Block is used for the inversion of the abundance maps. Experiments with both synthetic and real hyperspectral datasets confirmed the effectiveness and robustness of the proposed GSUU method, compared with the other typical unmixing algorithms. In this work, the number of endmembers was known as prior knowledge. In the future, the number of endmembers could be automatically estimated.

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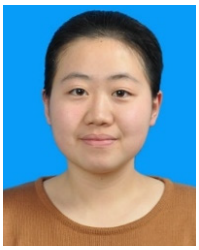
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